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TITLE PLACEHOLDER

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Abstract

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ABSTRACT PLACEHOLDER

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Introduction

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Suppose that you attend a friend’s wedding. That evening you describe it to your partner, re-
counting how the ceremony ran long, how the best man’s toast was unexpectedly moving, and
how a small child wandered onto the dance floor midway through the first song. A week later a
coworker asks how the wedding was, and you tell the story again. What is that second telling a
telling of? One appealing answer is that both tellings draw on the same underlying memory of
the wedding, and that whatever differences arise between them reflect the ordinary imprecision of
putting an experience into words. A second possibility, which motivates the work we report here,
is that your first telling has itself become part of what you are remembering.

14

What happens to an experience between living it and describing it later? Since our brains
do not maintain an exhaustive account of everything we encounter, the very act of perceiving an
experience already necessarily entails compressing and distorting it. We attend closely to some

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17 aspects of a scene while ignoring others, and we tend to interpret ambiguous moments in light
18 of what has happened previously and we expect will happen next. But once an experience has
19 ended, neither the experience itself nor our moment-to-moment perception of it remains available,
20 to us or to anyone else. What remains is instead the product of a further distillation, in which
21 our perception of the experience is reduced to something our memory systems can store and
22 later reconstruct. Which aspects of an experience survive this second distillation may depend on
23 what we attended to, what we anticipated would matter later, what we found interesting, and the
24 thoughts, motivations, and biases we brought with us. Because remembering entails reconstructing
25 an experience rather than replaying it, the resulting account often preserves the general “shape”
26 of what happened while discarding many of its particulars (Bartlett, 1932; Schacter, 2012; Schacter
27 et al., 2011). The units over which these distillations operate appear to be events, in that people
28 segment continuous experience into discrete episodes at moments of change, and those boundaries
29 shape what is remembered afterwards (Baldassano et al., 2017; Radvansky and Zacks, 2017). In
30 prior work we characterized this transformation geometrically by modeling an experience and a
31 person’s recounting of it as trajectories through a semantic space, and asking which properties of
32 the experience’s trajectory are preserved in the recounting (Heusser et al., 2021).

33 Yet memories do not remain fixed once they have been encoded, pointing to further transfor-
34 mations that a single recounting cannot reveal. Information may be lost over time; experiences we
35 have after an event may color how we later remember it; similar experiences may interfere with
36 one another; and we may simply re-evaluate what happened in light of what we have learned
37 since. The act of retrieving a memory itself appears to be one of the forces that changes it, in that
38 a memory can become labile when it is reactivated and must be stabilized again afterwards (Hup-
39 bach et al., 2007, 2009; Nader et al., 2000; Sederberg et al., 2011). Retrieval also changes what will
40 be accessible later—often substantially—relative to memories that go unretrieved (Karpicke and
41 Roediger, 2007; Roediger and Karpicke, 2006). Further, the particular way in which we recount
42 an experience appears to matter as well, since people who are asked to describe an event for one
43 purpose later remember it in ways that are biased toward that description (Tversky and Marsh,
44 2000). Memories also become more semantic and less perceptually detailed, both with the passage

45 of time and with repeated remembering (Lifanov et al., 2021). Repeated retrieval of naturalistic
46 events over the course of a week has been found to stabilize each person’s recall (for example,
47 which details they include) without causing different people’s recalls to converge (Melega et al.,
48 2025). Taken together, these observations suggest that each time we remember an experience, the
49 substance of the underlying memory we are retrieving may differ from what we retrieved before.

50 How might we compare two recountings of the same experience when the experience and
51 both recountings take the form of unconstrained natural language? Recent work has begun to use
52 language models to quantify the content of what individuals say when they recount what they
53 remember (Fenerci et al., 2025; Fitzpatrick et al., 2026; Heusser et al., 2021; Manning et al., 2022).
54 The approach developed in our prior work embeds the description of an experience, and each
55 individual’s recounting of it, into a common high-dimensional space whose coordinates reflect
56 the concepts under consideration, and through which the experience and recountings each unfold
57 over time as a geometric trajectory (Fitzpatrick et al., 2026; Heusser et al., 2021; Vera et al., 2025).
58 We then segment the experience and each recounting into events and identify which event from
59 the experience each recall event describes, yielding sequences of events that may be compared
60 directly across people and across delays (Baldassano et al., 2017; Chen et al., 2017; Heusser et al.,
61 2021).

62 Here we apply our approach to data from an experiment in which participants viewed an
63 episode of a television show, recounted its events immediately afterward, and then recounted
64 them again following a week-long delay. Because we collected two recountings of the same
65 experience from each participant, we are able to ask what participants’ delayed recountings most
66 closely resemble. If retrieval recovered the original experience in an uncorrupted form, then both
67 recountings should be about equally similar to the experience itself. We find instead that the
68 structure of a participant’s delayed recounting is more similar to that of their own immediate
69 recounting than to that of the original episode. It is also more similar to their own immediate
70 recounting than to other participants’ immediate recountings, to the degree that an individual’s
71 immediate recall can be identified from their delayed recall on the basis of this structure alone.
72 What governs retrieval after a delay therefore appears to be a person-specific distillation, rather

than the episode as it was first experienced.

Quantifying the semantic geometry of individuals' recalls at different points in time offers the opportunity to characterize the transformations undergone by our memories beyond an initial recounting. For example, if individuals' recountings tend to retain their idiosyncratic structures after a delay, how *do* the structures of our memories tend to change with time? There is a particular qualitative change our memories undergo as we naturally forget aspects of experience, implied by common descriptors such as "fading", "blurring", becoming "fuzzy", or "decaying" (Thorndike, 1913). But the nature of this change can be difficult to characterize more concretely. Here we show that this transformation is well described as geometrically simplifying the path through semantic space an individual follows as they recount an experience, in that applying existing algorithms for removing detail from a geometric curve to participants' immediate recalls identifies the details they are likely to forget following the delay.

Results

[OUTLINE — placeholder structure, not final prose. Bracketed italic text marks content still to be written. No statistic below has been computed or verified; every "report [STAT]" is a hole to be filled from the cited notebook's output.]

*[FIGURE BUDGET: 8 main-text figures maximum. The plan below uses 6, leaving 2 in reserve (one likely wanted for the RDP analysis if it needs a second panel set, one for contingency). Figure 1 (**paradigm.pdf**) is the only finished main-text figure; the other PDFs already in **paper/figures/source/** are analysis outputs that may feed panels rather than figures in their own right. Nothing else is made yet.]*

From episode and recalls to comparable event sequences

[Walk the reader through the pipeline, briefly and conceptually (details live in the Methods), up to the point where we have three things in hand: (a) the sequence of events comprising the episode, (b) each participant's sequences of immediate and delayed recall events, and (c) the mapping between each recall event and the

98 episode event it describes. This is the setup that everything below depends on — the point to land is that the
99 episode and both recalls now live in a common space and are segmented into comparable units.]

100 [Figure 1 — **paradigm**. `paradigm.pdf`, already in `paper/figures/source/`. Session-1 row: in-
101 structions, episode (24 min), immediate recall (10–60 min), prediction (5 min); one-week gap; session-2 row:
102 instructions, delayed recall (10–60 min), session-2 episode (20/22 min), recall of that episode (10–60 min).
103 Planned revision: reduce the opacity of the blocks for the phases we don’t analyze — the session-1 prediction
104 block, and the session-2 episode-viewing and subsequent-recall blocks — so the reader’s eye lands on the
105 immediate/delayed recall pair that the paper is actually about.]

106 [Figure 2 — **methods overview**. To be made. Annotations and recall text → trajectories → event-
107 segmentation procedure → event sequences. Loosely modeled on Fig. 2 of Heusser et al. (2021), extended
108 with the (new) event-segmentation step. Layout still open. Candidate source material already generated:
109 `episode-corrmat.pdf` and `k-optimization-episode.pdf` (from `event-segmentation.ipynb`), and
110 the per-recall similarity heatmaps and match matrices `atlep1-and delayed-recall-similarity-heatmaps.pdf`
111 `/ -matchmats.pdf`.]

112 **Participants recall the episode’s events in a similar order before and after the** 113 **delay**

114 [Motivation: we wanted to know whether participants remembered the episode similarly before and after the
115 delay. The simplest way to ask this is whether they recalled the same events in the same order.]

116 [Result: recall-order similarity. Computed in `self-similarity-analyses.ipynb`, section “Is de-
117 layed recall order more similar to own immediate recall order than to others’ immediate recall orders?”. The
118 measure compares pairwise precedence relations over the episode events shared between two recalls, then
119 percentile-ranks each participant’s delayed-vs-own-immediate similarity against their delayed-vs-others’-
120 immediate similarities, tested against chance (0.5) with a Wilcoxon signed-rank test. Report [STAT].]

121 [Figure 3 — **recall-order similarity**. To be made; multi-panel.]

122 [Panel A: illustration of the method. For one example participant, show the matrix of pairwise recall-
123 order-precedence values over pairs of episode events — probably just the lower triangle — for their delayed

124 recall beside their own immediate recall. Grey out (or reduce the opacity of) cells for event pairs not recalled
125 in both sessions, since the comparison is restricted to pairs present in both. Then show the same thing again
126 for that participant's delayed recall against a different participant's immediate recall, which is the contrast
127 the analysis actually tests. This maps directly onto `pairwise_precedence()` and the `shared_events`
128 argument to `order_similarity()` in the notebook, so the panel and the code should be checked against
129 each other.]

130 [Panels B–D: de-slopified versions of panels B–D of the reference mock-up. Include the analogues of that
131 mock-up's B and D so that the same panel types can reappear in the recall-structure figure (Figure 5), letting
132 the reader compare order and structure on identical terms — this parallelism is the reason for the pairing
133 and should be preserved in the final layout.]

134 [Note on ordering: this is the least novel analysis of the three and the one least dependent on our
135 framework — order could be measured without any embedding model. It goes first precisely because it
136 establishes the basic effect on the simplest possible terms, setting up the contrast with what follows.]

137 **The semantic structure of participants' recalls is preserved across the delay**

138 [Motivation: order is only the coarsest description of a recall. Because our approach embeds recalls in a
139 continuous semantic space, we can ask a stronger question — whether the structure of how a participant
140 moved through the episode's content is preserved, not merely the sequence in which they hit the events.]

141 [Figure 4 (optional) — 2D trajectories. Decision deferred, and it depends on how this section's
142 narrative settles. Argument for including it: it introduces the idea of geometric comparison and establishes
143 the 2D paths before they get used as examples in Figure 5, so the reader isn't meeting the representation
144 and the result at the same time. Argument against: it spends one of 8 figure slots on setup. Middle
145 option — a compact version showing only the episode path plus immediate and delayed paths for a handful
146 of participants, with the full version relegated to the supplement. Source: `2d-trajectories.ipynb`
147 (`atlep1-paths2d.pdf` and `delayed-paths2d.pdf`, `imm-del-2d.pdf`); note those `savefig` calls are
148 currently commented out in the notebook.]

149 [Result: trajectory/event-structure similarity. Computed in `self-similarity-analyses.ipynb`,
150 sections "Is delayed recall more similar to own immediate recall than to episode?" and "Is delayed recall

151 *more similar to own immediate recall than to others' immediate recalls?". Both use DTW warp cost (average*
152 *cosine distance between matched events) over mean-centered event sequences; the second percentile-ranks*
153 *self against others and tests against chance. Report [STAT] for each.]*

154 *[Framing: this result appears strong enough to present as a "fingerprint" — i.e., that one can identify*
155 *which immediate recall belongs to a given participant from the structure of their delayed recall alone. Frame*
156 *it that way in both the text and the figure, but confirm the effect size supports the claim before committing*
157 *to that language.]*

158 **[Figure 5 — recall-structure similarity.** *To be made; the largest figure in the paper, covering both*
159 *analysis (A) "delayed is more similar to immediate than to the episode" and analysis (B) "delayed is more*
160 *similar to own immediate than to others' immediates". Layout still open.]*

161 *[Example trajectory panels (3, from the current draft): the left panel is common to both analyses, so*
162 *it should sit centrally in the final arrangement; the right panel is the contrast for analysis (A) and the*
163 *middle panel is the contrast for analysis (B). Planned refinements: thicker lines; faint connectors between*
164 *DTW-matched events within each panel (already present in the draft); possibly PCHIP splines rather than*
165 *straight segments between points, to make the paths read as curves. Under consideration — a fourth panel*
166 *showing pairwise mean point-by-point distances between trajectories, which would require upsampling each*
167 *trajectory to a common number of points and comparing them in order (or substituting DTW cost, if that*
168 *turns out to read better).]*

169 *[Analysis (A) panel: per-participant connected dots showing each participant's delayed-to-immediate*
170 *versus delayed-to-episode alignment cost. Probably without the KDEs from the draft version.]*

171 *[Analysis (B) panels (3): de-slopified versions of the current draft. Two of them are the direct analogues*
172 *of Figure 3's panels B and D and show stronger effects — keeping the panel types visually identical across*
173 *the two figures is what makes "structure beats order" legible without the reader doing arithmetic. The third*
174 *gets at how person-specific the similarity is, and is the panel that carries the "fingerprint" claim.]*

175 *[Candidate supporting material: `atlep1-recall-corrstats.pdf` and `delayed-recall-corrstats.pdf`*
176 *(from `event-segmentation.ipynb`).]*

Delayed recalls are simplified versions of participants' immediate recalls

[Motivation: having established that a participant's delayed recall structure resembles their own immediate recall structure, the natural next question is how it changes over the delay.]

[Result: trajectory simplification. Computed in `rdp-analyses.ipynb`. Each participant's immediate recall trajectory is simplified with the Ramer-Douglas-Peucker algorithm over a range of tolerances (ϵ), and the ϵ minimizing DTW warp cost to their delayed recall trajectory is selected. Sub-results to report: (i) the relationship between how many events a participant dropped across the delay and how many RDP removes at the best-fitting ϵ (Kendall's τ); (ii) the same against ϵ itself; (iii) whether RDP predicts which events were forgotten, benchmarked against a hypergeometric null and compared to predictions derived from other participants' trajectories. Report [STAT] for each.]

[Figure 6 — RDP simplification. To be made, and how best to illustrate the simplification is genuinely unresolved — flagged as an open design question rather than a layout still to be drawn. The regression plots for sub-results (i) and (ii) exist in `rdp-analyses.ipynb` but have no `savefig` calls, so nothing is exported to `paper/figures/source` yet. One thing worth trying: since Figure 5 will already have taught the reader to read 2D trajectories, showing one participant's immediate path beside its RDP-simplified version and their actual delayed path would make “the delayed recall looks like a simplified version of the immediate one” visible directly, with the quantitative panels as support. If the analysis needs a second panel set, one of the 2 reserve figure slots can absorb it.]

[@paxton: does the simplification story want a fourth subsection on what the dropped events have in common (e.g. are they lower-salience, more peripheral to the narrative), or is that a Discussion point?]

Discussion

We asked participants to watch an episode of a television show, to recount what they remembered immediately afterwards, and to recount the episode again following a week-long delay. Embedding the episode's annotations and each recounting into a common semantic space allowed us to represent all three as trajectories, to segment each trajectory into events, and to compare them on common terms. Participants recalled the episode's events in a similar order before and after the

203 delay. The semantic structure of a participant's delayed recall was more similar to the structure
204 of their own immediate recall than to that of the episode, and it was more similar to their own
205 immediate recall than to other participants' immediate recalls. Delayed recalls were also well
206 described as simplified versions of immediate recalls, in that removing detail from a participant's
207 immediate recall trajectory brought it closer to the delayed recall trajectory that followed it.

208 Taken together, these results suggest a way of thinking about what recounting an experience
209 does to it. If experiences and memories may be described as trajectories through a semantic
210 space, then a recounting is a traversal of one of those trajectories (Heusser et al., 2021). Semantic
211 space differs from physical space in that nothing prevents a narrator from jumping between distant
212 locations, and yet participants' recountings turn out to be strongly autocorrelated in practice, which
213 is the regularity that our event-segmentation approach relies on. Because movement through
214 the space is relatively continuous, describing how one arrives at a later event from an earlier
215 one implies that the intervening region was traversed in some sense, whether it was articulated
216 explicitly or left for the listener to fill in. Some events therefore behave like control points, in that
217 removing one substantively changes the shape of the path. Others behave like waypoints, which
218 fall close to the line between their neighbors and may be removed with little consequence, since
219 travelling between the surrounding control points already implies passing near them. Trajectory
220 simplification provides a way of telling the two apart, and this distinction appears to hold at both
221 of the transitions we can examine. The events that simplification removes from the episode's own
222 trajectory are the events that the fewest participants recounted immediately afterwards (**[VERIFY**
223 **THIS]**), and the events that simplification removes from a participant's immediate recall trajectory
224 are the ones that participant tended to drop across the delay.

225 That not all events are equally memorable is well established. Gist and central details, for
226 example, survive delays that perceptual and peripheral details do not (Gilboa and Moscovitch,
227 2021; Heusser et al., 2021; Sekeres et al., 2016, 2018), and autobiographical narratives shed some
228 classes of content faster than others (Diamond and Levine, 2020; Su et al., 2026). Memories also
229 become more semantic and less perceptually detailed with the passage of time and with repeated
230 remembering (Lifanov et al., 2021), and events with stronger connections to the events around them

are better remembered than more isolated ones (Antony et al., 2024; Lee and Chen, 2022). What our results add is a change in where that asymmetry comes from. Centrality is ordinarily established from the outside, by asking raters which details matter to the plot, or by sorting content into categories fixed in advance. Here it follows from the geometry of an individual's own trajectory, and it is computable from that trajectory alone. An event is central because of where it sits relative to the rest of the path, and not because a rater judged it to be important. This turns the observation that some events are more memorable than others into a candidate explanation of why.

Why should waypoints be the events that go? One possibility is that they are redundant, in the sense that a listener, or the rememberer's own future self, can recover them from the events on either side. Another is that they are simply weaker in themselves, for example less distinctive, less emotionally charged, or less causally connected to the rest of the narrative (Antony et al., 2024). These accounts make similar predictions about which events will be forgotten, but they differ in what those predictions should depend on. If events are dropped because they are intrinsically weak, then a simplification derived from any participant's trajectory should predict forgetting about as well as one derived from the participant's own, since weak events would be weak for everyone. If they are dropped because they are recoverable from their neighbors, then a participant's own trajectory should do better, since control-point status is defined relative to the rest of that person's path. Our results favor the latter account. Read this way, dropping a waypoint is an economy rather than a loss, and it is worth noting that events which go unreported are not always unavailable, since a shift in perspective at retrieval can recover details that an earlier recounting left out (Anderson and Pichert, 1978; Oedekoven et al., 2017).

The same argument implies that what counts as a control point should be idiosyncratic. An event's status depends on the path it sits within, and different people who watched the same episode construct different paths through it, so the events whose removal would deform one person's trajectory need not be the events that matter for another's. This is what we take the specificity of the delayed recalls to mean. That a delayed recall can be matched to the immediate recall which preceded it, on the basis of structure alone, is a fingerprint in the sense that the term is used for the neural signatures by which individuals may be identified from functional connectivity or from

259 their responses to naturalistic stimuli (Anderson et al., 2020; Finn et al., 2020, 2015). The parallel
260 lies in the identification logic rather than in the measurement, since ours is a behavioral signature,
261 recovered from what a participant said rather than from how their brain responded. People differ,
262 for example, in where they place event boundaries in a continuous experience (Sava-Segal et al.,
263 2023), in how much of their recall structure they share with others (Chen et al., 2017; Nguyen
264 et al., 2019), and in the character of their autobiographical remembering more generally (Palombo
265 et al., 2018). Repeated retrieval of naturalistic events has been found to stabilize each person's
266 recall without causing different people's recalls to converge (Melega et al., 2025), which is what
267 one would expect if each person is stabilizing their own set of control points.

268 Everything we measured came from a verbal recounting, and so a reasonable objection is that
269 control points and waypoints are properties of storytelling rather than of memory. We think that
270 dichotomy is less clean than it first appears. Recounting is the only observable available in a study
271 of this kind, so the two are confounded by construction, and no analysis of transcripts alone can
272 fully separate them. More to the point, the way a person tells a story is not merely a window onto
273 what they remember, but also an input to what they will remember next. Retelling an event for a
274 particular audience or purpose, for example, biases how it is recalled later (Marsh, 2007; Tversky
275 and Marsh, 2000), and reproducing a narrative repeatedly reshapes it in ways that persist (Bergman
276 and Roediger, 1999; St Jacques and Schacter, 2013). If the function of recounting is to let a listener,
277 or one's own future self, re-traverse the trajectory of the original experience, then communicative
278 efficiency and mnemonic efficiency are not competing explanations so much as the same pressure
279 seen from two sides.

280 We acknowledge several limitations in the current study. First, every participant recalled the
281 episode immediately, and so we cannot separate two accounts of what we observed. A delayed
282 recall may resemble the immediate recall because the immediate recall was itself re-encoded and
283 then retrieved a week later, or because a single stable memory was sampled twice and each
284 sampling was shaped by the same person-specific distillation. Our measures are symmetric and
285 cannot establish direction, and the design lacks an arm in which the immediate recall is withheld.
286 Designs that do include such a control, in which retrieved information is later found nearer

287 to what was retrieved than to what was originally presented, offer the strongest evidence that
288 retrieval writes its own output back into the trace (Bridge and Paller, 2012; Nader and Hardt, 2009;
289 Nader et al., 2000). Second, we used one episode, one delay, and one intervening recounting, and
290 it remains to be seen how the picture changes across stimuli, across longer intervals, or across
291 many successive retellings (Chan et al., 2009; Lifanov et al., 2021). Third, control-point status is
292 defined with respect to one particular simplification algorithm, and a different algorithm might
293 well partition the same events differently. Fourth, our procedure matches every recall event to
294 an event in the episode, so content that a participant introduced but that does not appear in the
295 episode falls outside what we are able to measure.

296 Speculatively, this account suggests what might become of a memory that is recounted many
297 times. If each retelling preserves control points and sheds waypoints, and if each retelling then
298 becomes the trajectory that the next one simplifies, a memory might converge over many retellings
299 toward a skeleton of control points, with the intervening detail increasingly implied rather than
300 stated. Whether that convergence has a floor, and whether the surviving skeleton continues to
301 resemble the original experience or drifts steadily away from it, are questions our data cannot
302 answer. The possibility seems worth taking seriously outside the laboratory, since the settings
303 in which people are asked to recount an experience repeatedly are often the settings in which
304 accuracy matters most. An eyewitness who has already described what they saw, for example, is
305 describing a description the next time they are asked (Chan et al., 2009). A patient recounting a
306 formative episode across successive sessions may be reshaping it in the process, and a teacher who
307 explains an idea the same way each term may be simplifying it past the point where it still carries
308 what made it worth explaining.

309 Ultimately, our work points toward a set of questions about how the geometry of a recounting
310 relates to the experience that it describes. Do the events that behave like control points in a
311 recounting correspond to the boundaries at which neural activity shifts during the experience
312 itself (Baldassano et al., 2017; Chen et al., 2017)? Can they be predicted in advance from the
313 stimulus, or are they recoverable only after the fact, from a particular person's path? Does a listener
314 who hears a recounting actually re-traverse the speaker's trajectory, filling in the waypoints that

315 the speaker left out? And because our measure of what connects one event to another is built from
316 semantic similarity, it cannot on its own distinguish events that are close in meaning from events
317 that are causally linked (Antony et al., 2024; Lee and Chen, 2022; Xu et al., 2024). Collecting causal
318 and importance judgments for this episode would separate geometric control-point status from
319 causal centrality, and would tell us whether these two notions of what makes an event matter are
320 one notion arrived at by different routes.

321 **Methods**

322 Our experimental protocol was approved by the Committee for the Protection of Human Subjects
323 at Dartmouth College. We obtained written informed consent from all participants prior to the
324 start of both experiment sessions.

325 **Participants**

326 We enrolled a total of 70 Dartmouth undergraduate students in our study. Participants received
327 optional course credit for enrolling and were pre-screened to ensure they had not previously seen
328 any episode of either TV show viewed during the experiment. No statistical method was used
329 to predetermine sample size. Prior to data processing and analysis, we excluded a total of 17
330 participants from our dataset: 4 encountered technical issues with the experiment, 2 self-reported
331 difficulty with the task due to illness, 4 spoke for less than 10 minutes during the immediate recall
332 phase, and 7 did not return for the second experiment session. The data we analyzed comprised
333 the remaining 53 participants.

334 At the start of the experiment, we asked each participant to complete a demographic survey
335 that included questions about their age, sex, race, and ethnicity. Participants' ages ranged from 18
336 to 22 years (mean: 18.91 years; standard deviation: 1.01 years). A total of 41 participants reported
337 their sex as female and 12 participants reported their sex as male. Participants reported their races
338 as White (34 participants), Asian (16 participants), Black or African American (3 participants),
339 American Indian or Alaska Native (1 participant), and other (1 participant). (Note that some

participants selected multiple racial categories.) A total of 52 participants reported their ethnicity as “Not Hispanic or Latino” and 1 participant reported their ethnicity as “Hispanic or Latino.”

We also asked participants their native spoken language; whether they had any hearing, speech, or color vision impairments; and about any current medications or recent injuries that could impact attention or memory. All participants reported that their native spoken language was English, that they had no hearing or speech impairments, and that they had normal color vision. A total of 52 participants reported no attention- or memory-relevant medications or injuries. One participant reported sustaining a concussion more than 3 years prior to the experiment and was not excluded from our dataset.

Additionally, before each of the two experiment sessions, we asked participants about their sleep, coffee consumption, and level of alertness. Participants reported having slept between 5 and 11 hours the night prior to session 1 (mean: 7.37 hours; standard deviation: 1.45 hours) and between 5 and 11 hours the night prior to session 2 (mean: 7.39 hours; standard deviation: 1.23 hours). They reported having consumed between 0 and 3 cups of coffee on the day of session 1 (mean: 0.43 cups; standard deviation: 0.72 cups) and between 0 and 3 cups of coffee on the day of session 2 (mean: 0.48 cups; standard deviation: 0.68 cups). Participants reported their current levels of alertness on a 5-point Likert scale. At the start of session 1, 1 participant reported feeling “Very sluggish,” 14 reported feeling “A little sluggish,” 18 reported feeling “Neutral,” 15 reported feeling “A little alert,” and 5 reported feeling “Very alert.” At the start of session 2, 3 participants reported feeling “Very sluggish,” 9 reported feeling “A little sluggish,” 23 reported feeling “Neutral,” 17 reported feeling “A little alert,” and 3 reported feeling “Very alert.”

@jeremy: do we want to report class years and majors? And/or do we want to report post-experiment survey responses (how engaging did you find the episode, how easy/difficult was it to follow, how well do you think you recalled it, how well do you think you learned characters’ names)?

Experiment

The experiment took place over two sessions lasting approximately one hour each, separated by 6–8 days (mean: 6.76 days; standard deviation: 0.73 days). Prior to the start of the experiment,

participants were assigned to one of two groups (*A* or *B*), alternating based on enrollment order. During session 1, all participants first viewed the pilot episode of the FX series *Atlanta* (“The Big Bang”; duration: 24 min 15 s). Immediately after viewing, participants were given up to one hour to recount what happened in the episode to the best of their abilities. Following Chen et al. (2017), participants were instructed to describe the episode in as much detail as possible and to try to speak for a minimum of 10 minutes. They were told to try to recount the episode’s events in their original order, but that they were free to go back and describe earlier events should they realize they had missed something. After the immediate recall phase, participants were then given up to 5 minutes to predict what they thought would happen in the show’s next episode.

Upon returning for session 2, all participants were again given up to one hour to recount the *Atlanta* episode they had viewed the week prior. Following the delayed recall phase, participants in Group *A* then viewed the second episode of *Atlanta* (“Streets on Lock”; duration: 21 min 57 s) while participants in Group *B* instead viewed the pilot episode of the FOX series *Arrested Development* (“Pilot”; duration: 20 min 37 s). Both groups were then given up to one hour to recount the just-watched episode in the same manner and given the same instructions as in session 1. We implemented the experiment using the psiTurk platform (Gureckis et al., 2016) and jsPsych library (de Leeuw, 2015), with a custom jsPsych plugin we developed in prior work for recording verbal recall audio (Ziman et al., 2018). Here we analyze participants’ immediate and delayed recalls of the *Atlanta* episode viewed in session 1.

Stimulus annotation and recall transcription

To capture the evolving semantic content of the *Atlanta* episode, we created detailed text annotations of the on-screen content during each individual camera shot throughout the episode (mean annotation duration: 3.05 s; standard deviation: 1.80 s). Specifically, we used ANVIL (Kipp, 2001) to record, for each camera shot, a set of semantically relevant features. These included a brief description of physical occurrences and characters’ explicit actions, a brief description of characters’ internal states (e.g., emotions, motivations) and other implicit content, the names of all characters appearing on screen, a transcription of any dialogue, the name(s) of the character(s) speaking, the

394 shot’s setting, whether the shot took place indoors or outdoors, and whether or not there was
395 music playing during the shot. This feature set was adapted from a similar annotation corpus
396 created by Chen et al. (2017), which we used in developing our trajectory-analysis framework in
397 prior work (Heusser et al., 2021).

398 We then transcribed each participant’s immediate and delayed verbal recall of the *Atlanta*
399 episode. Our transcription process consisted of three steps: an initial manual transcription, a
400 two-pass prose-formatting conversion with a large language model, and a final manual cleanup
401 and correction step. First, we manually transcribed each recall audio file collected during the
402 experiment. These initial transcripts reflected participants’ spoken recalls verbatim, without any
403 punctuation or additional formatting, and including disfluencies, stuttered repetitions, and other
404 artifacts common to unrehearsed speech. However, since transformer models (such as we used
405 to obtain text embeddings of participants’ recalls) are trained almost exclusively on written text,
406 structural conventions like sentence breaks, punctuation, and capitalization provide an important
407 signal these models leverage to contextual other tokens (e.g., words) and construct a representation
408 of the text’s overall meaning. Meanwhile, speech-native phenomena such as stuttered repetitions
409 and filler words (“like”, “um”, etc.) that are comparatively rare in written text contribute out-of-
410 distribution noise that can be expected to degrade embedding fidelity, and prior work has shown
411 that automated removal of these phenomena improves performance on downstream tasks that
412 assume clean, written-style input (Chen et al., 2022; Wang et al., 2010). We therefore used a large
413 language model (Qwen3.5 122b; Qwen Team, 2026) to convert participants’ recall transcripts to
414 a structure and style more typical of written text. We used the `langchain_dartmouth` Python
415 package (Stone et al., 2025) to implement this conversion as two sequential passes over each
416 transcript. In the first pass, we prompted the model to split the transcript into sentences at
417 natural breaking points, add punctuation including quotes around reported speech, remove clear
418 non-word disfluencies (“um,” “uh,” “hmm,” etc.), collapse stuttered verbatim repetitions, and
419 mark self-corrections and restarts with em-dashes. In the second pass, we prompted the model
420 to remove all instances of “like” used as a filler word. We removed filler “like” instances in a
421 separate pass because of their high frequency of occurrence in the transcripts and the additional

422 effort to required to distinguish them from meaningful uses of “like.” The full texts of our prompts
423 for both passes can be found in *Supplementary methods*. Finally, we manually reviewed all LLM-
424 formatted transcripts to identify and correct any errors, ensure sentence breaks were appropriate
425 and consistent across participants, confirm that no content was improperly added to or removed
426 from the transcripts, and resolve any ambiguities based on the prosody of participants’ original
427 recall audio.

428 **Analysis**

429 **Statistics**

430 Wilcoxon tests were one-sided because our alternative hypothesis was directional (delayed matches
431 own immediate *better than* the episode, others’ immediates, etc.).

432 **Constructing episode and recall trajectories**

433 We adapted a method we developed in prior work (Heusser et al., 2021) to transform the *Atlanta*
434 episode and participant’s recalls of it into trajectories through a common representation space
435 using a text embedding model. Briefly, text embedding models take in natural language text
436 of an arbitrary length and output a fixed-length vector representation that captures the text’s
437 semantic meaning. Intuitively, an output vector of length n can be viewed as a coordinate in an
438 n -dimensional text embedding space. While specific model architectures and training paradigms
439 vary, modern text embedding models are broadly trained to output similar coordinate vectors (in
440 terms of cosine similarity, dot product, inverse Euclidean distance, or other geometric measures) for
441 semantically similar input texts, even when those texts share few or no specific words. We obtained
442 text embeddings for the episode and recalls using EmbeddingGemma (Vera et al., 2025), a pre-
443 trained 308-million-parameter encoder-only transformer model that outputs coordinate vectors in
444 a 768-dimensional embedding space.

445 To construct the trajectory for the *Atlanta* episode, we first converted our tabular annotations
446 of the content present during each camera shot to a plain-text format. Separately for each shot,

we concatenated the annotation text describing explicit on-screen occurrences and characters' implicit internal states. We then appended "Speech: " followed by the shot's dialogue, "Character speaking: " followed by the name of the speaker, and "Setting: " followed by the shot's setting. If any annotated fields were empty for a given shot (e.g., if no character was speaking at that time), we omitted them and their prefixes from the concatenated text. Next, we used EmbeddingGemma to transform the text snippet for each camera shot into a text embedding vector, resulting in a number-of-shots (478) by number-of-features (768) matrix. Since the camera shots varied in duration, we then resampled this series of embedding vectors to a consistent temporal resolution. We assigned each vector to the timepoint midway between the start and end of its corresponding camera shot (in seconds), and used linear interpolation to estimate a vector for each second of the episode. This yielded a number-of-seconds (1,455) by number-of-features (768) matrix—i.e., a 768-dimensional *trajectory*.

We then created similar trajectories for each participant's immediate and delayed recall of the episode. We began by splitting each recall transcript into a list of sentences, with the constraint that splits could not occur within reported speech (i.e., if a participant recounted a multi-sentence quote by a character in the episode, we treated the full quote as a single sentence-unit). We then parsed these sentences into overlapping sliding windows that spanned up to 5 consecutive sentences each. These windows grew to and shrank from their full size at the beginning and end of a participant's recall sequence—in other words, the first window contained only the first sentence of the participant's recall, the second contained the first two sentences, and so on. This ensured that all sentences from a participant's recall transcript appeared an equal number of times (5) in their sequence of sliding windows. Finally, we used EmbeddingGemma to transform each sliding window into a text embedding vector. Repeated for all recall transcripts, this yielded a separate trajectory characterizing each participant's recall from each of the two experiment sessions. The number of samples in participants' recall trajectories (i.e., the number of sliding windows parsed from their recall transcripts) ranged from 61 to 308 (mean: 136.68; standard deviation: 49.70).

473 Relating episode and recall embedding vectors

474 In prior work using our trajectory-analysis framework (e.g., Fitzpatrick et al., 2026; Heusser et al.,
475 2021; Manning, 2021; Manning et al., 2022; Racciah et al., 2024), embeddings of the stimulus and
476 participants’ responses were constructed using a topic model (Latent Dirichlet Allocation; Blei
477 et al., 2003) trained solely on the stimulus content. Here we elected to instead use a pre-trained
478 transformer-based text embedding model due to the richer and more nuanced textual information
479 these models are able to capture. (For example, these context-sensitive models can disambiguate
480 the texts “Earn looked at Alfred” and “Alfred looked at Earn” in their representations, whereas
481 bag-of-words models like LDA would represent the two identically.) However, these richer rep-
482 resentations also introduce certain geometric nuisances we needed to account for in order to
483 effectively relate the contents of participants’ recalls to that of the episode they viewed. First,
484 because modern transformer models are pre-trained on large and diverse text corpora, the repre-
485 sentational capacities of their embedding spaces span an enormous breadth of semantic content.
486 Consequently, in preliminary analyses we found that the embeddings of our annotations and par-
487 ticipants’ recalls (which were all “about” the events, themes, and characters of a single *Atlanta*
488 episode) were strongly anisotropic—i.e., they all occupied the same relatively narrow angular
489 region of the model’s overall embedding space. Additionally, because transformer models are sen-
490 sitive to the syntactic structure of text, their embedding representations carry latent information
491 about the text’s linguistic register in addition to its semantic content. Accordingly, we found there
492 was a consistent angular offset between embeddings of the episode annotations and those of par-
493 ticipants’ recalls, reflecting the fact that the annotations comprise grammatically well-structured
494 written text while the recalls comprise transcripts of casual unrehearsed speech.

495 We corrected for these representational artifacts by mean-centering the episode and recall trajec-
496 tories before performing analyses that involved measuring similarities between embedding vectors
497 within or across trajectories. Conceptually, mean-centering a given set of embedding vectors (i.e.,
498 subtracting the across-vector average from each individual vector) serves to remove representa-
499 tions that are common across all vectors under consideration, and instead highlight representations

that *vary* between them. For example, consider the sequence of embedding vectors comprising the trajectory of the *Atlanta* episode. Common across vectors within this sequence will be latent signals driven by how we chose to format the annotation text passed to the embedding model, the annotator’s particular idiolect (i.e., idiosyncratic vocabulary and writing style), and any high-level overarching themes of the episode. *Not* common across these vectors are the finer-grained semantic representations that distinguish different events, scenes, and actions throughout the episode from each other. We found that removing these non-discriminative signals (to the extent they behave as additive offsets) via mean-centering helped reduce the embeddings’ anisotropy (i.e., increased the range of measured between-vector similarities) and improved our ability to segment the episode trajectory into meaningful events. Similarly, mean-centering both the episode and recalls before relating their embedding representations helped mitigate broad-scale representational differences driven by their diverging registers, and enabled us to better relate them based on event-specific, content-driven patterns.

Our approach to mean-centering the recalls was guided by the particular latent representation we wanted to remove versus retain. In segmenting each participant’s recall trajectory from each session into events (see *Event-segmentation*), our goal was to discriminate between embedding vectors strictly within a single recall sequence in order to identify temporally transient content-related patterns. We therefore centered each recall trajectory (in addition to the episode) on its own centroid to remove non-discriminative representations (e.g., register, broad-scale thematic content) and maximize the contrast between vectors within that single sequence. However, the analyses shown in Figures **REF TO DTW FIG** and **REF TO RDP FIG** entailed comparing sequences of recall events not only to the episode, but also to each other, both within- and across-participants, to examine whether within-participant similarities tended to exceed across-participant similarities. Here our goal was to remove representational artifacts that could work to bias our results towards significance. Specifically, participants’ idiolects (i.e., particular speaking styles and sets of words with which they describe the episode) will influence the embedding representations of their recalls in ways that tend to be similar between their immediate and delayed recalls, but different across participants. To correct for this confound, we jointly centered each participant’s set of immediate

528 and delayed recall events on the two sequences’ shared centroid (i.e., the average vector across
529 all events from both sessions). This served to reduce the recall embeddings’ anisotropy (by
530 removing signals driven by high-level themes of the episode they recalled) and mitigate the effects
531 of the spoken-recall register and participant-specific idiolect (both of which are shared between a
532 participant’s recalls from the two sessions) while preserving any genuine drift in the high-level
533 content of each participant’s recall over the delay (which per-session mean-centering would instead
534 discard).

535 **Event-segmentation**

536 After constructing trajectories for the episode and each participant’s immediate and delayed recall,
537 we used hidden Markov models (HMMs; Baum and Petrie, 1966; Rabiner, 1989) to segment each
538 trajectory into a sequence of discrete events. Given a number of states K , an HMM models an
539 observed time series as a progression through K hidden states, each of which has a characteristic
540 signature in the data (i.e., a pattern that timepoints belonging to that state tend to exhibit) and
541 tends to persist across timepoints with occasional transitions rather than fluctuating freely. Because
542 states persist, inferring the most probable state at each timepoint segments the observed time series
543 into a sequence of events—i.e., spans of contiguous timepoints with the same most-probable state.

544 We segmented the episode trajectory into events using the HMM implemented in the BrainIAK
545 toolbox (Kumar et al., 2021), which jointly learns a mean pattern for each of K states and the set
546 of transitions between them via the Baum-Welch algorithm (Baum et al., 1970), with the added
547 constraint that each state is visited exactly once in a strict left-to-right order (Baldassano et al., 2017).
548 To select an appropriate number of states (i.e., events) for the episode, we used an optimization
549 procedure we developed in prior work (Heusser et al., 2021) that aims to identify the single
550 timescale of event representations emphasized most heavily in the episode’s narrative structure.
551 We began by segmenting the (mean-centered) episode trajectory into K events for each K between
552 2 and 50, inclusive. For each candidate segmentation, we then computed the cosine similarity
553 between each pair of timepoints assigned to the same event, and between each pair of timepoints
554 assigned to different events. When constructing these distributions of within- and across-event

555 similarities, we restricted comparisons to pairs of timepoints separated by no more than the
556 duration of the longest event in the candidate segmentation, so that the within- and across-
557 event distributions were computed over comparable temporal distances and the across-event
558 distribution was not dominated by distant timepoint pairs that we would expect to be dissimilar
559 regardless of event structure. We then selected the candidate segmentation that maximized the
560 Wasserstein distance between these within- and across-event similarity distributions. We found
561 that the maximally distinctive and internally self-similar segmentation was yielded by $K = 20$
562 events. **(TODO: ADD SUPP FIG SHOWING WASSERSTEIN DIST AS FUNCTION OF K)**
563 Finally, we averaged the embedding vectors over timepoints between each pair of adjacent event
564 boundaries to obtain a sequence of stable event-average representations for the episode.

565 To segment participants' recall trajectories into sequences of events, we developed a novel
566 HMM-driven procedure that grounded the recalls directly against the content of the episode
567 participants viewed and recounted. Essentially, rather than re-learning separate sets of state
568 representations from each participant's recalls (which tend to be shorter, noisier approximations
569 of the episode content), we leveraged the state representations already learned from the episode to
570 identify transitions between the episode events a given participant was (most probably) describing
571 at a given point in time. First, we mean-centered the sequence of event-average embedding
572 vectors for the episode, and each participant's sequence of per-window embedding vectors from
573 each session, separately. For each given recall trajectory, we then computed the cosine similarity
574 between the embedding vector for each recall window and the event-average embedding vector
575 for each of the 20 episode events. This yielded a number-of-windows by number-of-episode-
576 events (20) similarity matrix. We then treated the episode events as the hidden states of a hidden
577 Markov model with fixed parameters, and treated each recall window's profile of similarities
578 to the episode events, z-scored across all window-event pairs, as its distribution of emission
579 scores over states. We imposed an ergodic transition structure (i.e., we allowed transitions from
580 any state to any other state, since participants may recall the episode events out of order, and
581 allowed states to be re-entered multiple times, since participants may revisit earlier events in
582 their recalls) governed by separate additive transition penalties for forward-one transitions and

all other transitions. Specifically, we imposed no penalty for remaining in the current state, a small penalty (1.5 normalized emissions units) for transitioning from a given state to the next (i.e., transitioning from recalling episode event n to recalling episode event $n + 1$), and a larger penalty (3.0 normalized emissions units) for all other transitions. Finally, we decoded the most probable sequence of episode-event labels using the Viterbi algorithm (Viterbi, 1967), collapsed runs of consecutive identically labeled recall windows into events, and averaged the embedding vectors over windows within each event. This procedure yielded, for each participant’s recall from each session, a sequence of event-average embedding vectors and a set of labels denoting which episode event each recall event corresponded to.

Comparing trajectory structures

We compared the geometric structures of the episode and recall event-sequences using dynamic time warping (DTW; Sakoe and Chiba, 1978; Vintsyuk, 1968). The DTW algorithm finds the optimal order-preserving alignment between two sequences of vectors with potentially different lengths or sampling rates by locally “stretching” and “compressing” them in time to maximize their overall similarity. This alignment yields a “warp path” that maps each vector in each sequence onto one or more consecutive vectors in the other sequence with the minimum total distance between matched vectors. The summed or average distance between matched vectors along the warp path (the “warp cost”) is commonly taken as a measure of the two sequences’ structural dissimilarity. We computed the warp cost as the average cosine distance between DTW-matched events so that the cost values would be comparable across comparisons involving event-sequences of different lengths (whereas summed distance necessarily grows with sequence length).

Data availability

All of the data analyzed in this manuscript may be found at <https://github.com/ContextLab/memory-dynamics>.

Code availability

All of the code for running our experiment and carrying out the analyses may be found at <https://github.com/ContextLab/memory-dynamics>.

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